

Bernat Casajuana Suarez

[Email](#), [Website](#), [GitHub](#)

Biomedicine student. Applied mathematics and scientific machine learning for biological systems.

Education

BSc in Biomedicine, University of Vic – Central University of Catalonia	2022 – Present
Erasmus Exchange, Università degli Studi di Milano	2025 – 2026
BSc in Biochemistry (60 ECTS), Autonomous University of Barcelona	2021 – 2022

Research Interests

My research interests lie at the interface of applied mathematics, scientific machine learning, and biology. I work on the modeling and analysis of complex dynamical systems and networks, combining mechanistic approaches with data-driven methods for inference, control, and prediction. I aim to develop rigorous quantitative frameworks that bridge mathematical theory and biological mechanism, with applications to biomedical phenomena.

Research Experience

Computational and Systems Biology	2025 – Present
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[Computational Biochemistry and Biophysics Lab](#), UVic-UCC

- Reverse engineering of biological dynamical systems using Physics-Informed Neural Networks (PINNs)
- Parameter inference and trajectory recovery for the repressilator model from noisy and partially observed time-series data

Work Experience

Mathematics Private Tutor	2024 – 2025
Waiter, Hotel - Restaurant L'AVET	Summer 2024
Laboratory Assistant, Avinent Implant System	Summer 2023

Technical Skills

Programming: Python, R

Scientific computing & machine learning: NumPy, SciPy, DeepXDE, TensorFlow

Developer tools: Git, LaTeX

Transferable Skills

Structured approach to complex problem solving

Clear communication of scientific results

Quick adaptation to new research frameworks and technical environments

Languages

Catalan and Spanish (native)

Fluent in English (B2 certified)

Basic understanding of Italian